

PayGuard + ClaimGuard — Demo Runbook

One Payment-Integrity platform: post-pay recovery (PayGuard) and pre-pay review (ClaimGuard).

Created 2026-08-15 · verified live + fixes 2026-08-17 · findings tiering + cross-check added 2026-08-24 | PRODUCTION |

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1. What this demo shows

Two complementary payment-integrity pipelines, same suite, same top-nav app switcher:

- **PayGuard (post-pay)** — claims already paid; 23 detectors have flagged overpayments into a prioritized worklist the analyst works to recovery.
- **ClaimGuard (pre-pay)** — upload a scanned claim, then attach the medical record, *before* payment; the AI audits the claim and checks the record for the evidence each diagnosis needs.

Arc: lead with PayGuard (“here’s what we already recovered”), then ClaimGuard (“now watch us stop it *before* the check goes out”). Both apps at penguinai.studio; switch via the top-right nav.

2. Part 1 — PayGuard walkthrough (post-pay)

Step A — Dashboard (executive hook)

Open `payguard.penguinai.studio`: 29 open cases, 12 high-priority, ~\$22.6K at risk, status/aging/detector-performance panels.

Step B — Review worklist

`/worklist?stage=review` — 14 prioritized cases, filterable by LOB / detector / assignee; show detector breadth.

Step C — Walk one case

Recommended: OPA-2026-00020 (Stacy Truman, DET-18 Medical Necessity, \$258.33) — not overdue; intuitive story. **Alt:** OPA-2026-00009 (Ostrowski, DET-09 AI Coding Errors, \$1,386) — bigger + shows AI, but currently OVERDUE.

Walk: detector results → evidence → AI “Suggested: Recoup it (99%)” → decision → recoupment letter → recovery. End on the case’s **Documentation / Evidence tab** (“Attach a medical-record PDF”) as the bridge into ClaimGuard.

3. Part 2 — ClaimGuard walkthrough (pre-pay)

Verified end-to-end with the Reyes files. This is a full sibling app, not just an upload button.

3.1 — Orientation

`claimguard.penguinai.studio` (“ClaimGuard AI — Payer Claims Review”). Own **Dashboard** (Claims Under Review, \$ Billed Under Review, Prevented Overpayment MTD, aging) and **queues**: All Claims / Pending / In Review / Needs Info / Escalated / Closed, plus Team Monitor, Reports, Members, Admin.

3.2 — Upload the claim (“New Claim from PDF”)

All Claims → **New Claim from PDF** → drop `reyes_cms1500.pdf`. Takes ONE claim PDF (CMS-1500/UB-04). A 4-step “Parsing your claim...” loader runs ~15-20s (upload → read form → pull fields → create claim + run AI audit).

Extraction is accurate: Mildred Reyes, Midwest Cardiac, DOS Apr 2026, **billed \$1,250**, CPT **99223 / 93306 / 93000**, ICD I50.21 / I10. Click **Open claim**.

3.3 — The AI audit (Claim Details + AI Findings)

Top banner: a recommendation (Approve / **Manual review** / Reject) with an evidence %, per-line charges, and a **Generate Denial Letter** button. Two rounds of fixes shipped to production land the audit on the *real* issue instead of a wall of noise:

Finding	Status after fixes
FWA-03 Place-of-Service: 99223 (initial <i>hospital</i> E/M) billed POS 11 (office); should be POS 21/31.	KEEPS — the HERO. Correct upcoding catch; feature it.
FWA-02 Specialty: notes 99223 inconsistent with the outpatient setting.	Keeps — now carries a cross-check note distinguishing it from FWA-03 (see 3.5).
DET-13 Code Validity: I50.21 “not in reference tables.”	REMOVED — I50.21 added to the ICD seed.
DET-18 Med-Necessity: echo 93306 / ECG 93000 “not medically necessary.”	REMOVED — heart-failure dx now covered (echo/ECG indicated for acute systolic HF).
DET-18 0%-confidence “no coverage rules” card for 99223.	REMOVED — DET-18 skips E/M visit codes.

The climax to tell (Manual review): *“The AI surfaced the one real problem — a hospital-admission E/M code billed as an office visit — and routed the claim to a human rather than auto-denying. It did NOT deny the echo or ECG, which the record shows are medically appropriate. That’s a payer that recovers what it should without over-denying legitimate care.”* Lead with FWA-03; the per-finding Accept/Reject buttons make human-in-the-loop the point.

3.4 — Evidence Check: the “why you need both documents” climax

This is the strongest ClaimGuard moment. The **Evidence Check** tab lists, per diagnosis, what a medical record must contain to substantiate it.

	Before medical record	After attaching reyes_encounter_notes.pdf + Re-scan
Result	2 missing / 0 found	2 found / 0 missing
I50.21	“Gap: no medical-record documents attached.” Requires dyspnea/edema, echo or BNP, systolic-vs-diastolic note.	Evidence found · high — quotes the note: “...LVEF of 30% with new NYHA Class III symptoms, elevated NT-proBNP... support an acute systolic HF exacerbation (I50.21).” Cited to p.1 / ASSESSMENT, with View in PDF.

Two-step flow to rehearse: (1) upload the claim — Evidence Check shows 2 *missing*; (2) go to the **Documents** tab, attach reyes_encounter_notes.pdf, return to Evidence Check, click **Re-scan** — the diagnoses flip to **found** with page-level citations. That visible flip *is* the “both scanned, useful together” punchline.

3.5 — Key issues vs Other signals: findings tiering + cross-check

Shipped 2026-08-24 to keep the AI Findings tab from reading as “this claim is littered” when many rules fire. Two behaviors to point out:

- **Tiering.** Findings split into a primary **Key issues** list (top 10 by materiality) and a collapsible **Other signals** group. A one-line summary states the full count (e.g. “14 findings fired; showing the 10 most material; 4 in Other signals”) — nothing is hidden, just prioritized.
- **Only material findings drive the recommendation.** Below-threshold flags ($\leq 50\%$ confidence, unless a high-\$ / high-severity safety-valve keeps them up) move to Other signals and do NOT inflate the evidence % — so the headline reconciles with what’s shown.

- **Cross-check distinction notes.** One holistic LLM pass over the Key set adds a one-sentence note to any two findings that overlap. On Reyes this is the payoff for the “FWA-02 and FWA-03 look like duplicates” reaction: each now carries a sky-blue note — e.g. FWA-03 reads “unlike FWA-02 (which flags the provider’s specialty), this concerns the place-of-service code.”

What to say: “When lots of rules fire, we don’t bury the reviewer — we rank the material issues, tuck the low-confidence noise into Other signals, and where two findings overlap the AI explains the difference in one line. Full transparency, no false ‘littered claim’ impression.” Reyes (2-3 findings) shows the cross-check notes; to show the Key/Other split live you need a claim that trips many rules (e.g. the Ostrowski \$38.5K UB-04).

4. The Reyes documents — why claim + record beat either alone

File	test_files/reyes_cms1500.pdf	test_files/reyes_encounter_notes.pdf
Says	POS 11 = Office, but CPT 99223 = “Initial HOSPITAL E/M, high complexity”; total \$1,250	Visit Type: Office / Outpatient; CHF exacerbation, LVEF 30%, NT-proBNP 6,200, IV furosemide
Together	99223 is a hospital-admission code billed in an office — wrong setting / upcoded E/M (caught by FWA-03).	The record proves the office setting AND substantiates I50.21/I10 (Evidence Check flips to found).

One-liner: “The scanned claim tells us *what was billed*; the scanned record tells us *what actually happened* — only reading both surfaces the wrong-setting E/M *and* substantiates the diagnoses.” Member MA-000007 and provider Midwest Cardiac are seeded, so intake accepts them.

5. Test-file reference (repo)

PDFs at ./test_files/ unless noted. Intake rejects unseeded members/providers — stick to these.

reyes_cms1500.pdf + reyes_encounter_notes.pdf	Claim + record pair, cardiac (CHF)	THE ClaimGuard demo (verified)
Stacy_Truman_Ortho_Note_20241205.pdf (repo root)	Ortho note	Pairs with PayGuard OPA-2026-00020
john_doe_readmission_MR.pdf	248KB dense inpatient CRBSI readmission	Showcase extraction depth
Ostrowski UB-04 (\$38.5K, in the seeded queue)	Many-line institutional claim	Show the Key/Other tiering split live

6. Reset / reseed before the demo

- **No reseed needed to run the demo** — uploads create fresh claims live.
- **Railway is ephemeral** (no volume; SEED_ON_EMPTY=1): every redeploy wipes the DB, re-migrates, and auto-seeds. **A redeploy is what activates the shipped fixes + the tiering/cross-check feature** — both repos (izeinoun/opa backend and izeinoun/ClaimGuard frontend) must have deployed.
- **Redeploy to:** (1) clear rehearsal clutter (any practice uploads) — they vanish on redeploy; (2) re-anchor stale dates (most PayGuard cases otherwise show OVERDUE).
- **Destructive + your action:** a redeploy deletes everything not in the seed, and Claude cannot trigger it.

Recommended: redeploy once for fresh dates + clean slate, rehearse, then don’t redeploy again right before going live.

7. Pre-demo checklist

- [] Redeploy BOTH repos so the fixes + tiering/cross-check are live (else the old noisy findings still show).
- [] PayGuard: Dashboard loads; feature case pre-opened (OPA-2026-00020, or Ostrowski if redeployed).
- [] ClaimGuard: have reyes_cms1500.pdf (claim) + reyes_encounter_notes.pdf (record) ready.
- [] Dry-run: upload claim → recommendation = **Manual review**, led by FWA-03; DET-13 / DET-18 noise gone; FWA-02/03 show cross-check distinction notes → attach record → Re-scan → Evidence Check flips to 2 found.
- [] (Optional) open a many-finding claim (Ostrowski) to show the Key issues / Other signals split + summary line.
- [] Rehearse the “Manual review” line and the “we rank, we don’t bury” tiering line.

Verified live 2026-08-17; tiering/cross-check shipped 2026-08-24. Case IDs, dollar amounts, and deadlines reflect the seeded production dataset and shift on the next reseed. Members/providers named in test files must exist in the seed or intake returns 422.